

Assignment No. 2

Procedure

Procedure (Step-by-Step)

- Step 1: Update System
- Step 2: Install KVM and Required Packages
- Step 3: Verify Installation

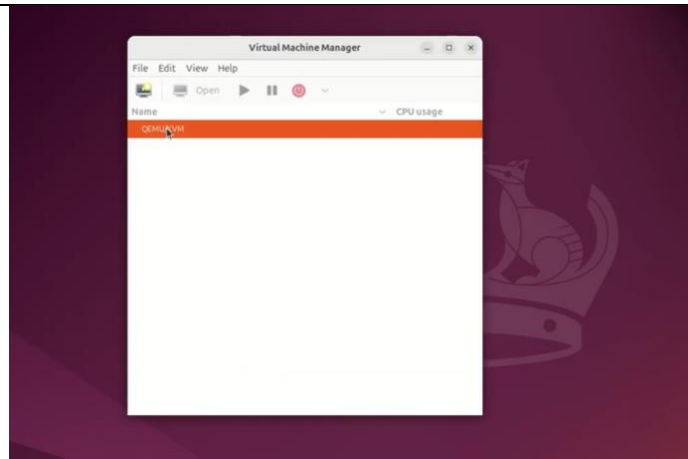
```
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo apt install cpu-checker
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvml7t64 python3-netifaces
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  msr-tools
The following NEW packages will be installed:
  cpu-checker msr-tools
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 15.8 kB of archives.
After this operation, 65.5 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 msr-tools amd64 1.3-5build1 [9,610 B]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/main amd64 cpu-checker amd64 0.7-1.3build2 [6,148 B]
Fetched 15.8 kB in 1s (13.1 kB/s)
Selecting previously unselected package msr-tools.
(Reading database ... 150839 files and directories currently installed.)
Preparing to unpack .../msr-tools_1.3-5build1_amd64.deb ...
Unpacking msr-tools (1.3-5build1) ...
Selecting previously unselected package cpu-checker.
Preparing to unpack .../cpu-checker_0.7-1.3build2_amd64.deb ...
Unpacking cpu-checker (0.7-1.3build2) ...
Setting up msr-tools (1.3-5build1) ...
Setting up cpu-checker (0.7-1.3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ kvm -ok
Command 'kvm' not found, but can be installed with:
sudo apt install qemu-system-x86
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo apt install qemu-kvm virt-manager libvirt-daemon-system libvirt-clients bridge-utils -y
Reading package lists... Done
Building dependency tree... 50%
```

- Step 4: Start and Enable Services
- Step 5: Add User to libvirt Group

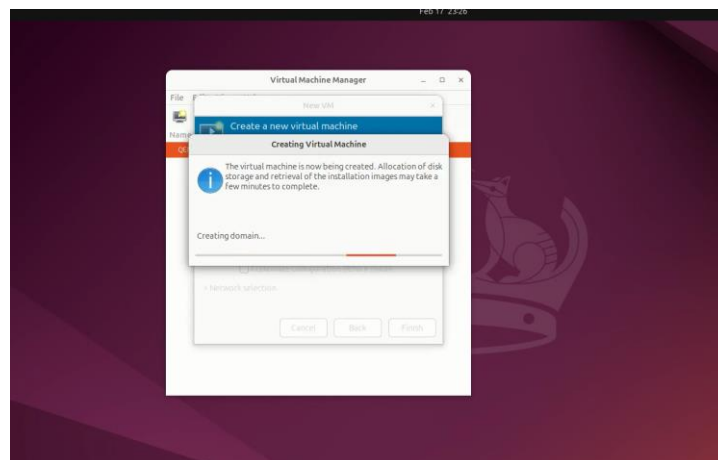
```
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo systemctl enable libvirtd
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start libvirtd
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo usermod -aG kvm kishore
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo usermod -aG libvirt kishore
kishore@kishore-VMware-Virtual-Platform:~/Desktop$ sudo systemctl status libvirtd
○ libvirtd.service - libvirt legacy monolithic daemon
   Loaded: loaded (/usr/lib/systemd/system/libvirtd.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2025-02-17 23:23:53 IST; 16s ago
     Duration: 2min 97ms
   TriggeredBy: ● libvirtd-admin.socket
                 ● libvirtd-ro.socket
                 ● libvirtd.socket
   Docs: man:libvirtd(8)
          https://libvirt.org/
   Process: 5734 ExecStart=/usr/sbin/libvirtd $LIBVIRT_ARGS (code=exited, status=0/SUCCESS)
   Main PID: 5734 (code=exited, status=0/SUCCESS)
     Tasks: 2 (limit: 32768)
    Memory: 5.5M (peak: 13.0M)
       CPU: 626ms
   CGroup: /system.slice/libvirtd.service
           └─5843 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
             5844 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper

Feb 17 23:21:53 kishore-VMware-Virtual-Platform dnsmasq[5843]: using nameserver 127.0.0.53#53
Feb 17 23:21:53 kishore-VMware-Virtual-Platform dnsmasq[5843]: read /etc/hosts - 8 names
Feb 17 23:21:53 kishore-VMware-Virtual-Platform dnsmasq[5843]: read /var/lib/libvirt/dnsmasq/default.addnhosts - 0 names
Feb 17 23:21:53 kishore-VMware-Virtual-Platform dnsmasq-dhcp[5843]: read /var/lib/libvirt/dnsmasq/default.hostsfile
Feb 17 23:21:53 kishore-VMware-Virtual-Platform libvirtd[5734]: libvirt version: 10.0.0, package: 10.0.0-2ubuntu8.5 (Ubuntu)
Feb 17 23:21:53 kishore-VMware-Virtual-Platform libvirtd[5734]: hostname: kishore-VMware-Virtual-Platform
Feb 17 23:21:53 kishore-VMware-Virtual-Platform libvirtd[5734]: Unable to open /dev/kvm: No such file or directory
Feb 17 23:23:53 kishore-VMware-Virtual-Platform systemd[1]: libvirtd.service: Deactivated successfully.
Feb 17 23:23:53 kishore-VMware-Virtual-Platform systemd[1]: libvirtd.service: Unit process 5843 (dnsmasq) remains running after unit stopped.
Feb 17 23:23:53 kishore-VMware-Virtual-Platform systemd[1]: libvirtd.service: Unit process 5844 (dnsmasq) remains running after unit stopped.
kishore@kishore-VMware-Virtual-Platform:~/Desktop$
```

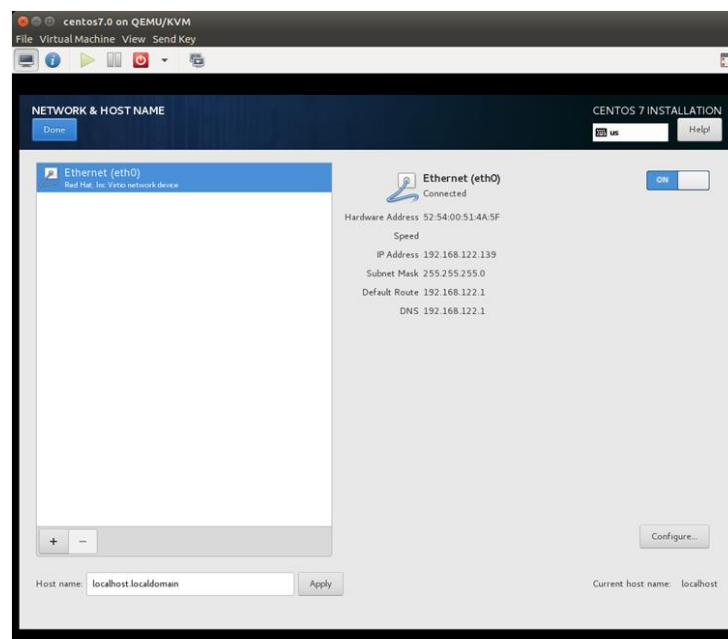
- Step 6: Launch Virtual Machine Manager



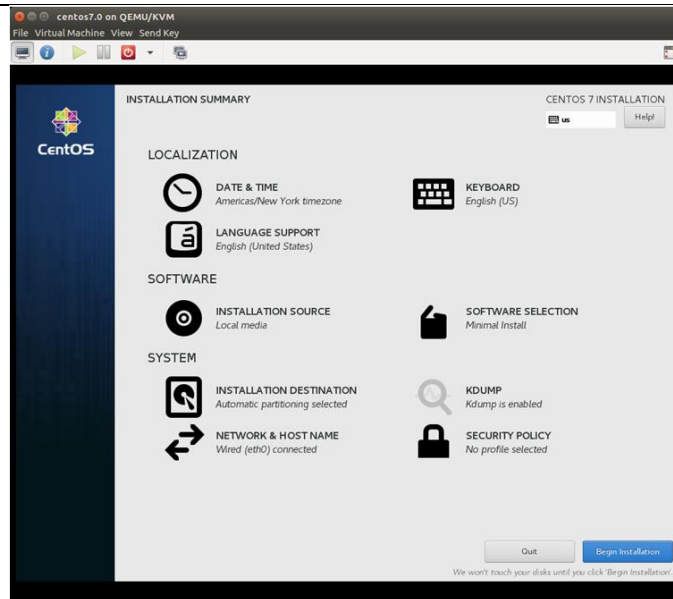
- Step 7: Create a Virtual Machine



- Step 8: Configure Networking



- Step 9: Manage Virtual Machine



Observations and Results

- The system successfully detected hardware virtualization support.
- KVM packages were installed without errors.
- The `kvm-ok` command confirmed that KVM is enabled.
- The `libvirtd` service was running successfully.
- Virtual Machine Manager launched correctly.
- A virtual machine was created and the OS installation completed successfully.
- The VM was able to boot and run independently.

Analysis

The practical shows how Kernel-based Virtual Machine uses hardware virtualization to run multiple virtual machines efficiently on a single system. It provides high performance with low overhead due to its integration with the Linux kernel. Overall, it is a scalable and cost-effective virtualization solution.

Conclusion

The practical demonstrated the installation and configuration of KVM as a virtualization solution. It showed how a Linux system can be converted into a hypervisor to run multiple virtual machines efficiently. KVM provides a powerful, open-source, and scalable virtualization platform suitable for cloud and enterprise environments.